Learning-Based Flexible Cross-Layer Optimization for Ultrareliable and Low-Latency Applications in IoT Scenarios

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***Abstract*—With the continuous popularization and deepening of the Internet-of-Things (IoT) technologies, trillions of IoT Devices (IoTD) are connected to the network. The huge growth of wireless communication traffic and the surge of energy consumption make it a great challenge to support various requirements of IoTDs, such as ultrareliable and low latency. The 6th-generation (6G) network has put forward new goals and visions for green communication, network flexibility and intelligence, which are expected to solve these key challenges. In this article, we propose a cross-layer optimization scheme to achieve the trade-off between energy efficiency (EE) and spectral efficiency (SE) of the 6G enabled IoT networks, where the ultrareliable and low-latency applications are considered. Flexible self-organization of three parameters is realized, namely, transmission time interval (TTI), packet duplication (PD), and resource block (RB) allocation. The key technology of flexible TTI scheduling guarantees the reduction of latency, and the PD transmission can effectively improve the reliability. Furthermore, based on machine learning (ML) method, we propose the transfer asynchronous advantage actor–critic (TA3C) algorithm to realize parameter configuration and resource allocation. The simulation results show that the EE and SE tradeoff performance of our proposed flexible scheme is improved by at least 39.29% compared with the fixed parameter configuration. In addition, the TA3C algorithm has better convergence performance and reduces the algorithm complexity by up to 91.23% compared with other ML algorithms.**

***Index Terms*—6G, delay, Internet of Things (IoT), intelligence, packet duplication (PD), reliability, transmission time interval (TTI).**

Manuscript received 21 October 2020; revised 5 March 2021; accepted 17 April 2021. Date of publication 28 April 2021; date of current version 8 August 2022. This work was supported in part by National Key R&D Program of China under Grant 2020YFB1806905; in part by the National Natural Science Foundation of China under Grant 61871045; and in part by the BUPT Excellent Ph.D. Students Foundation under Grant CX2020203. *(Corresponding author: Xiaodong Xu.)*

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Digital Object Identifier 10.1109/JIOT.2021.3076230

I. Introduction

With the launching of the 5th-generation (5G) wireless networks, studies of the 6th-generation (6G) wireless networks are ongoing, which aims at achieving the intelligent information society enabled by fully dimensional wireless coverage, green communication, inspired intelligence and flexible configuration. Many scholars look forward to the development visions of 6G from various aspects and dimensions [1], [2].

Internet of Things (IoT) is viewed as the key enabler in the convergence of physical space and virtual space world [3]. Trillions of IoT Devices (IoTD) will be connected to the 6G networks and provide communication, control, computation, and consciousness services [1], [2]. However, it is full of challenges to support massive IoTDs with various requirements in 6G networks, such as extremely low latency, ultrahigh reliability, etc. How to achieve ultrareliable and low-latency communications (uRLLC) in mission critical IoT scenarios becomes a critical issue.

Guaranteeing the flexibility of 6G network is important to overcome the aforementioned challenges [4]. In the 6G network, the scheduling of resources will be more fine-grained, the network configuration will be more flexible [5]. Some reforms of physical-layer and frame structure, including multinumerology and mini-slot, etc., bring important opportunities to meet the requirements in 6G enabled IoT networks.

Because of the sharp increase of energy consumption results from the rapid growth of communication traffic, the solutions of energy-saving communication are imminent [6]. Green communication has become an important goal and vision of 6G network development [7]. The main metrics that researchers pay attention to include energy efficiency (EE), spectrum efficiency (SE), power consumption, etc. These factors play vital roles in the 6G network especially for the energy-limited devices, such as sensors and various kinds of connected things.

Furthermore, it is expected that 6G will support ubiquitous artificial intelligence (AI) services. The intelligence will be one of the kernel force for future communication network performance improvement [8]. In recent years, machine learning (ML) has been widely used in uRLLC scenarios, providing a greater possibility and broader prospect for solving the problems that difficult to model and solve [9].

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Although the vision of 6G network has been vividly described, due to the complexity of the network and the diversity of application requirements, the related technologies and researches are still in the initial stage. 6G networks will provide customized services for more applications, which makes it more advantageous in complex energy consumption, flexible configuration, network intelligence, and so on [10].

The existing communication networks are layered and most technologies are completed independently, but the interaction between different layers cannot be ignored [11]. In order to meet the delay and reliability requirements of IoTDs, it is necessary to research a cross-layer optimization scheme. Therefore, in view of the above key issues, we focus on the delay and reliability sensitive IoTDs in the 6G network, and propose a cross-layer optimization scheme. The main contributions are as follows.

1) We propose a cross-layer optimization scheme based on flexible parameter configuration and resource allocation, which improves the trade-off performance of EE and SE by more than 39.29% compared with the traditional fixed configuration scheme. The trade-off between EE and SE is taken as the optimization goal, in line with the green communication vision of 6G network.
2) The self-configuration and self-optimization of three network parameters are realized, namely transmission time interval (TTI), packet duplication (PD) number, and resource block (RB) allocation. Flexible TTI scheduling is used to reduce the delay, and PD transmission can effectively improve the reliability. Customized services are provided for IoTDs with different requirements, which improves the flexibility of 6G network.
3) We introduce the knowledge of transfer learning (TL) and make full use of the historical learning experiences. The proposed transfer asynchronous advantage actor–critic (TA3C) algorithm has stable convergence performance, and effectively reduces the complexity of the algorithm up to 91.23%, which increases the intelligence of the network.
4) The stochastic network calculus (SNC) is adopted as the modeling method, and ML is combined to confront the disadvantage of SNC. Our method avoids the approximation of SNC, and can express the network performance more accurately.

The remaining of this article is arranged as follows. In Section II, the related works about flexible configuration, green communication and ML are listed. The cross-layer system model and SNC method are introduced in Section III. Section IV shows the optimization problem we considered in the above background. In Section V, solution of cross-layer optimization method based on ML is detailed. Section VI introduces the specific simulation scheme and results. Finally, there are concluding remarks.

II. Related Works

Many researchers have focused on the resource distribution, security, access and many other aspects of the IoT scenarios [12]–[15]. Chettri and Bera [12] introduced the challenges and opportunities IoT faced, and provided some promising application fields and research directions. It points out that delay, low cost, SE, and EE are the key words in IoT scenarios. Unlike most of the existing papers focusing on the security and privacy, Xing [13] indicated that the reliability of IoT should be defined as the reliable operation of key tasks within the specified time. A multistate sequential learning framework with limited memory is given in [14], which makes the IoTDs allocate their limited communication resources in a self-organization way. Chen *et al.* [15] utilized SNC to analyze the network latency, proposed a improved K-means algorithm to cluster the narrowband IoT terminals, and recommended the priority-based scheduling strategy in different clusters.

5G new radio technology makes it possible for flexible network architecture and physical-layer structure [16]. Zhang *et al.* [17] pointed out that flexible TTI scheduling is necessary for delay-sensitive services. Flexible numerology in time domain and frequency domain is studied in [18], and a suboptimal resource allocation scheme is proposed. Although the advantages of flexible TTI scheduling have been proved in some works [18], [19], there are few studies on specific TTI selection and configuration methods. There are also lots of studies on how to improve the reliability of services in the network. Rao and Vrzic [20] showed that PD is an economical and effective method to improve transmission reliability without significant modification of network deployment. Without increasing the operation complexity, PD can meet the reliability requirements. Elayoubi *et al.* [21] focused on the industrial IoT, and study the resource allocation scheme under the retransmission mechanism to meet the reliability requirements with low resource consumption. To the best of our knowledge, there is no work combining the flexible TTI and PD technologies to meet the strict delay and reliability requirements of services.

ML has been gradually applied to uRLLC scenarios [22]–[25]. Due to the strict delay requirement of uRLLC, the solution of problem needs to be implemented within one TTI. Compared with the traditional optimization methods, ML algorithms can effectively reduce the complexity and processing delay. In addition, You *et al.* [9] pointed out that the problems, which are difficult to model or solve, are important application prospects of ML in the future researches. ML has advantages of solving complex scenarios and nonconvex problems. What is more, how to improve the ML methods for uRLLC scenarios is another issue. Based on federated learning, a distributed method is proposed in [23], which enables the joint transmit power and resource allocation for ultrareliable and low-latency vehicular communication. From a comprehensive perspective, She *et al.* [11] expounded the application challenges of the uRLLC in 6G network, put forward the integration of communication area knowledge into deep learning network, and pointed out it is meaningful to research cross-layer optimization strategies.

Therefore, we propose a cross-layer optimization scheme based on flexible TTI and PD transmission. The improved ML method is applied to realize the flexible configuration of network parameters, and the delay and reliability requirements of service are met.

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TABLE I
LIST OF MAIN PARAMETERS

Notations	Meanings	Notations	Meanings
k	IoT device	m	Basic station
$\mathcal{U}_{\mathcal{K}}$	The set of device	$\mathcal{B}_{\mathcal{K}}$	The set of BS
$T^{f}_{k,m}$	TTI size	$T^{c}_{k,m}$	The time of control overhead in one slot
$T^{t}_{k,m}$	The time of data transmission in one slot	$N^{RE}_{k,m}$	The number of REs for control overhead
N^{RE}_{RB}	The number of REs contained in one RB	$W_{k,m}$	The number of subcarriers
$P_{k,m}$	Transmit power	$r_{k,m}$	Transmission rate
$B_{k,m}$	Bandwidth of subcarrier	α	Large-scale fading
$g_{k,m}$	Small-scale channel gain	N_0	Unilateral noise spectral density
ε_c	Decoding error probability	$Q_G(x)$	Gaussian Q-function
$Q_{k,m}$	The length of queue	$prob_{k,m}(\omega_{k,m})$	Delay violation probability
$L^{q}_{k,m}(t)$	Queueing delay	D_{max}	Delay requirement
$\varepsilon_{\max}$	Reliability requirement	V	Channel dispersion
$\mathcal{A}_{k,m}, \mathcal{S}_{k,m}, \mathcal{D}_{k,m}$	The processes in bit domain	$\mathcal{M}_{\mathcal{A}_{k,m}}, \mathcal{M}_{\mathcal{S}_{k,m}}, \mathcal{M}_{\mathcal{D}_{k,m}}$	The processes in SNR domain
$\varepsilon^{q}_{k,m}$	Delay violation probability	μ_k	Packet size
η_{EE}	Energy Efficiency	η_{SE}	Spectrum Efficiency
η^{norm}_{EE}	Normalized EE	η^{norm}_{SE}	Normalized SE
$\eta^{\max}_{EE}$	Maximum of EE	$\eta^{\max}_{EE}$	Maximum of SE
N_{BS}	The number of BS	W_{BS}	The bandwidth of BS
$N^{PD}_{k,m}$	The number of PD	ϑ	The preference for EE
$t_{k,m}$	The transmission time	σ_c, σ_a	Learning rate of actor and critic

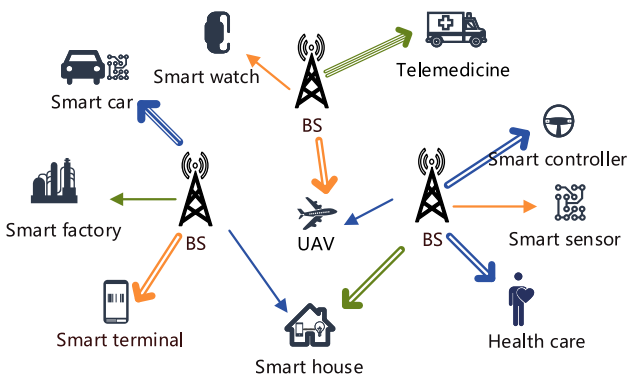

Fig. 1. System model. The yellow arrows indicate that the TTI size are 0.0625 ms, the blue arrows indicate that the TTI size are 0.125 ms, and the other green arrows indicate that the TTI size are 0.25 ms. Overlapping arrows represent the PD transmission.

III. SYSTEM MODEL

In this article, we consider the 6G enabled IoT communication scenario, where several basic stations (BSs) and IoTDs with extremely strict requirements of delay and reliability are presented. Some devices are shown as examples in Fig. 1, such as smart car, telemedicine, etc. The BSs implement the configuration of the TTI size, the number of PD transmission, and RB parameter based on experience and different requirements of IoTDs. Customized services are provided based on optimized parameters. The important parameters in this article are shown in Table I.

To meet the strict QoS requirements of uRLLC services, the strong interferences should be avoided. Indeed, the intracell interferences can be solved by orthogonal multiple access techniques, and the intercell interferences can be avoided by interference coordination technologies, such as using the 1/3 as the intercell frequency reuse factor [26]. The remaining interferences in the network are weak and could be treated as noise [27].

A. Physical-Layer Model

5G new radio and numerology technology standardized by the 3rd Generation Partnership Project (3GPP) have provided support for flexible frame structure and multiple deployment options [28]. According to the definition of multinumerology, the time domain and frequency domain of the frame structure are determined when the relevant parameters are determined. When the time domain of RB becomes smaller, the frequency domain will be multiplied accordingly. It has been widely proved that shorter TTI has performance advantages for delay sensitive services [18]. Therefore, according to the 3GPP standards and multinumerology technology, we consider 0.0625, 0.125, and 0.25 ms for flexible selection in this article. As shown in Fig. 2, the green rectangle represents the case of TTI = 0.25 ms. When TTI becomes 0.125 ms, the frequency domain is doubled, and the RB structure is shown in yellow rectangle. The more detailed description is given in our previous work [29]. Different RB structures coexist in the system, which can improve the utilization efficiency of communication resource and meet the requirements of IoTDs.

We consider a discrete-time model in duration T, in which time is divided into time slots. Each time slot is divided into control signal transmission and data transmission as illustrated in Fig. 2. We use k represents the IoTD, use m represents the BS, and $k \in \mathcal{U}_{\mathcal{K}}, m \in \mathcal{B}_{\mathcal{K}}$. Define the slot of k connected to m as $T^{f}_{k,m}$, the time occupied by the control overhead transmission as $T^{c}_{k,m}$, and the time occupied by data transmission

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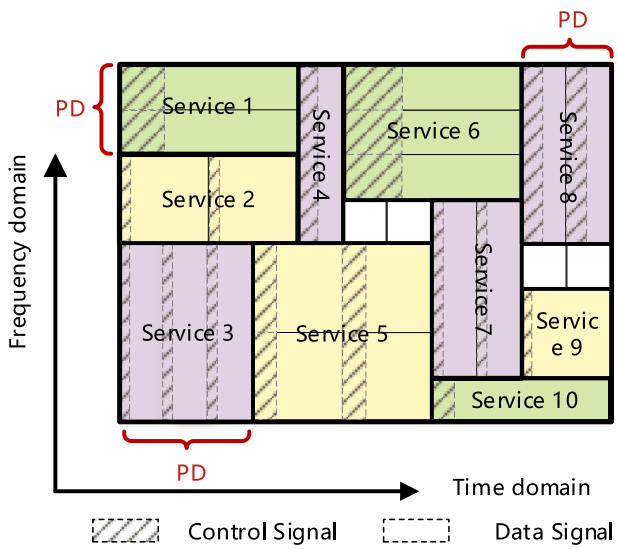

Fig. 2. Illustration of resource allocation with flexible TTI, PD, and RB configuration.

as $T^t_{k,m}$. Then, there are

$$T^c_{k,m} + T^t_{k,m} = T^f_{k,m}. \tag{1}$$

The size of control overhead is related to the signal-to-noise ratio (SNR) of the wireless transmission channel. When the channel quality deteriorates, the control overhead increases. On each scheduling opportunity, corresponding amount of resource element (RE) must be occupied to transmit the control signaling. The shaded part in Fig. 2 represents control signaling transmission, and the rest part is data transmission. $N^{\mathrm{RE}}_{k,m}$ represents the number of REs required for control overhead. The following equation shows the relationship between control overhead and SNR [30]:

$$N^{\mathrm{RE}}_{k,m} = \left\{ \begin{array}{ll} 36, & \mathrm{SNR} \in [4.2, +\infty) \\ 72, & \mathrm{SNR} \in [0.2, 4.2) \\ 144, & \mathrm{SNR} \in [-2.2, 0.2) \\ 288, & \mathrm{SNR} \in (-\infty, -2.2) \end{array} \right\}. \tag{2}$$

The number of REs contained in one RB is $N^{\mathrm{RE}}_{\mathrm{RB}} = 168$. Furthermore, the relationship between the $T^c_{k,m}$ and $T^f_{k,m}$ can be represented as

$$T^c_{k,m} = \frac{N^{\mathrm{RE}}_{k,m}/N^{\mathrm{RE}}_{\mathrm{RB}}}{W_{k,m}} \cdot T^f_{k,m} \tag{3}$$

where $W_{k,m}$ is the number of subcarriers allocated by the BS m to the IoTD k. The BS can select the specific length of TTI for the service according to the expriences and the characteristics of the service.

To guarantee the reliability requirements of IoTDs, the PD transmission technology is taken into account. When the reliability requirement of IoTD is strict, or the channel quality is poor, PD can significantly improve the reliability to meet the requirements of IoTDs. In this article, the number of PD, expressed with $N^{\mathrm{PD}}_{k,m}$, can be selected among 0, 1, 2, and 3. $N^{\mathrm{PD}}_{k,m} = 0$ means no duplication is required. Considering that the short TTI takes up more bandwidth and the transmission is faster in the time domain, the duplication of short TTI will be completed flexibly in the time domain. Due to the short transmission time, the duplication in time domain can still meet the delay requirements of IoTDs, which will be proved in our simulation results. As shown in Fig. 2, the flexible choice of TTI and PD effectively allocates the 2-D resources in time domain and frequency domain, which improves the resource utilization of the network.

B. Wireless Transmission Model

A flat Rayleigh block-fading channel model is considered, in which the channel remains constant for the duration of one slot and changes independently between slots. To support service for delay sensitive IoTDs, the data packets designed for these devices are very small, and only channel codes with finite blocklength can be used [26]. The Shannon theory based on infinite blocklength cannot capture the transmission characteristics of delay sensitive IoTDs with short packets [31]. In short packet transmission, the transmitted blocklength and transmission error probability cannot be ignored. Based on the finite blocklength coding theory [26], we approximate the achievable rate of delay sensitive IoTD as

$$r_{k,m} \approx \frac{B_{k,m}W_{k,m}}{\ln 2} \cdot \left[\ln\left(1 + \frac{\alpha g_{k,m}P_{k,m}}{N_0 B_{k,m}W_{k,m}}\right) - \frac{\sqrt{V_{k,m}}Q_G^{-1}(\varepsilon_c)}{\sqrt{T^t_{k,m}B_{k,m}W_{k,m}}} \right] \tag{4}$$

where $P_{k,m}$ is the transmit power allocated by BS m to IoTD k, $B_{k,m}$ is the bandwidth of subcarrier, which is related to the length of TTI. α is the large-scale fading determined by the path loss and shadow fading. The small-scale channel gain is denoted by $g_{k,m}$ and it obeys the Rayleigh distribution. N_0 is the unilateral noise spectral density, ε_c is the decoding error probability, and $Q_G(x)$ is the Gaussian Q-function

$$Q_G(x) = \int_x^{\infty} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz \tag{5}$$

the channel dispersion is represented by $V_{k,m}$

$$\sqrt{V_{k,m}} = \sqrt{1 - \frac{1}{\left(1 + \frac{\alpha g_{k,m}P_{k,m}}{N_0 B_{k,m}W_{k,m}}\right)^2}}. \tag{6}$$

Due to the complicated calculation of $V_{k,m}$, it often leads to a nonconvex optimization problem, and brings a great challenge to the analyses with traditional optimization methods. Most of the related works simplify (6), or approximate it to the fixed value 1 [26]. Because we use ML method instead of traditional optimization analysis, this approximation can be avoided. We adopt the precise form of (6) to further guarantee the accuracy of calculation.

C. Queueing Model

SNC is a modeling method of network layer and can be used to analyze delay performance, especially for delay sensitive services. The delay violation probability derived from SNC combines the delay and reliability closely. The SNC method

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is based on a suitable traffic model and service model, and focuses on the upper boundary of cumulative arrival of service and the lower boundary of cumulative service provision.

We focus on the arrival, service and departure processes of the buffer at the BS. In the duration (τ, t), we define the cumulative data arrival, system cumulative service and cumulative data departure of IoTD k to BS m as

$$\begin{aligned} A_{k,m}(\tau, t) &= \sum_{i=\tau}^{t-1} a_{k,m}(i) \\ S_{k,m}(\tau, t) &= \sum_{i=\tau}^{t-1} s_{k,m}(i) \\ D_{k,m}(\tau, t) &= \sum_{i=\tau}^{t-1} d_{k,m}(i) \end{aligned} \tag{7}$$

where $a_{k,m}$ is the data bits arrive at the buffer, $s_{k,m}$ represents the amount of service the system provides to IoTDs, which can be expressed by the transmission process represented by (4). The amount of departure is represented by $d_{k,m}$, which is defined as the number of bits successfully received by the IoTD. The amount of departure $d_{k,m}$ is related to the queuing length at the current moment, arrival process and system service capacity. $Q_{k,m}(i)$ represents the queuing length of IoTD k to BS m at time i, then

$$d_{k,m}(i) = \min\{Q_{k,m}(i) + a_{k,m}(i), s_{k,m}(i)\} \tag{8}$$

$$Q_{k,m}(i+1) = Q_{k,m}(i) + a_{k,m}(i) - d_{k,m}(i). \tag{9}$$

Considering the principle of first-come–first-served (FCFS), the queuing delay of the IoTD at the BS at time t is expressed as

$$L^q_{k,m}(t) = \inf\{u \geq 0 : A_{k,m}(0, t) \leq D_{k,m}(0, t+u)\}. \tag{10}$$

Furthermore, we define the probability that the delay at time t exceeds the maximum expected $D_{\max}$ as the delay violation probability $\text{prob}_{k,m}(D^k_{\max})$. This indicator can be used as an important parameter for analyzing reliability of service

$$\text{prob}_{k,m}\left(D^k_{\max}\right) = \sup_{t\geq 0}\left\{\Pr\left[L^q_{k,m}(t) > D^k_{\max}\right]\right\}. \tag{11}$$

Because the closed-form solution of the delay violation probability is difficult to obtain, we approach the upper limit of this probability through SNC. Furthermore, we transform the analyses in bit domain to exponential domain, or in other words, SNR domain [32]. Then, the arrival, service and departure processes can be expressed as

$$\begin{aligned} \mathcal{A}_{k,m}(\tau, t) &= e^{A_{k,m}(\tau,t)} \\ \mathcal{S}_{k,m}(\tau, t) &= e^{S_{k,m}(\tau,t)} \\ \mathcal{D}_{k,m}(\tau, t) &= e^{D_{k,m}(\tau,t)}. \end{aligned} \tag{12}$$

We use $\oslash$ represents deconvolution operator

$$\alpha \oslash \beta(\tau, t) = \sup_{u\leq\tau}\left\{\frac{\alpha(u, t)}{\beta(u, \tau)}\right\}. \tag{13}$$

The queuing delay in (10) can be rewritten as

$$L^q_{k,m}(t) = \inf\{u \geq 0 : \mathcal{A}_{k,m} \oslash \mathcal{S}_{k,m}(t+u, t) \leq 1\} \tag{14}$$

The probability of delay violation is further expressed as [32]

$$\begin{aligned} \text{prob}_{k,m}\left(D^k_{\max}\right) &= \Pr\left[L^q_{k,m}(t) > D^k_{\max}\right] \\ &\leq \Pr\left[\mathcal{A}_{k,m} \oslash \mathcal{S}_{k,m}\left(t + D^k_{\max}, t\right) \leq 1\right] \\ &\leq \mathcal{M}_{\mathcal{A}_{k,m}\oslash\mathcal{S}_{k,m}}\left(1+s, t + D^k_{\max}, t\right) \\ &\leq \inf_{s>0}\left\{\mathcal{K}_{k,m}\left(s, D^k_{\max}\right)\right\} \end{aligned} \tag{15}$$

where $\mathcal{M}_X(s, \tau, t) = E[X(\tau, t)^{s-1}]$ is the Mellin transform for any $s \in R$ when the expectation of a non-negative random process $X(\tau, t)$ exists, $E(\cdot)$ represents the expectation. $\mathcal{K}_{k,m}(s, D^k_{\max})$ is the steady-state kernel function satisfying (10)

$$\begin{aligned} \mathcal{K}_{k,m}\left(s, D^k_{\max}\right) = \lim_{t\to\infty}\sum_{u=0}^{t} &\mathcal{M}_{\mathcal{A}_{k,m}}(1+s, u, t)\mathcal{M}_{\mathcal{S}_{k,m}} \\ &\times \left(1-s, u, t + D^k_{\max}\right). \end{aligned} \tag{16}$$

The Mellin forms of the arrival process and service process in the SNR domain are expressed as

$$\begin{aligned} \mathcal{M}_{\mathcal{A}_{k,m}}(s, \tau, t) &= E\left\{\left[\prod_{i=\tau}^{t-1} e^{a_{k,m}(i)}\right]^{s-1}\right\} \\ &= \left\{E\left[e^{a_{k,m}(s-1)}\right]\right\}^{t-\tau} \\ &= \left[\mathcal{M}_{\xi_{k,m}}(s)\right]^{t-\tau} \end{aligned} \tag{17}$$

$$\begin{aligned} \mathcal{M}_{\mathcal{S}_{k,m}}(s, \tau, t) &= E\left\{\left[\prod_{i=\tau}^{t-1} e^{r_{k,m}(i)}\right]^{s-1}\right\} \\ &= \left\{E\left[e^{r_{k,m}(s-1)}\right]\right\}^{t-\tau} \\ &= \left[\mathcal{M}_{\psi_{k,m}}(s)\right]^{t-\tau} \end{aligned} \tag{18}$$

where $\xi_{k,m} = e^{a_{k,m}}$, $\psi_{k,m} = e^{r_{k,m}}$. Substitute (17) and (18) into (16)

$$\mathcal{K}_{k,m}\left(s, D^k_{\max}\right) = \frac{\left[\mathcal{M}_{\psi_{k,m}}(1-s)\right]^{D^k_{\max}}}{1 - \mathcal{M}_{\xi_{k,m}}(1+s)\mathcal{M}_{\psi_{k,m}}(1-s)}. \tag{19}$$

Formula (19) should satisfy the constraints of $\mathcal{M}_{\xi_{k,m}}(1+s)\mathcal{M}_{\psi_{k,m}}(1-s) < 1$. We notice that if we can obtain the Mellin form of the arrival and service process, we can deduce the upper bound of the delay violation probability.

IV. Cross-Layer Optimization Scheme

A. Combination of SNC and Finite Blocklength Coding

In this section, we combine SNC and finite blocklength coding to obtain an approximate form of the delay violation probability. We express the number of packets arrive at the buffer as λ_k in each slot, and the packet size as μ_k bits. The Mellin transformation of the arrival process is expressed as [32]

$$\mathcal{M}_{\xi_{k,m}}(s) = \sum_{n=0}^{\infty} e^{n(s-1)}\frac{(\mu_k\lambda_k)^n}{n!}e^{-\mu_k\lambda_k} = e^{\mu_k\lambda_k\left(e^{s-1}-1\right)} \tag{20}$$

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Meanwhile, we rewritten (4) as

$$r_{k,m} = \frac{B_{k,m}W_{k,m}}{\ln 2}\left[\ln\left(1 + Jg_{k,m}\right) - \sqrt{V_{k,m}}N\right] = B_{k,m}W_{k,m}\log_2\frac{\left(1 + Jg_{k,m}\right)}{e^{\sqrt{V_{k,m}}N}}. \tag{22}$$

The service process is calculated by the finite blocklength transmission theory expressed in (4). In order to express the calculation more concisely, we do substitutions as follows:

$$J = \frac{\alpha P_{k,m}}{N_0 B_{k,m}W_{k,m}}, N = \sqrt{\frac{1}{T^t_{k,m}B_{k,m}W_{k,m}}}Q_G^{-1}(\varepsilon_c). \tag{23}$$

We model the transmission process as the service process. The accuracy of transmission is $(1 - \varepsilon_c)$. The service process is expressed as [33]

$$s_{k,m}\left(g_{k,m}, \varepsilon_c\right) = \begin{cases} B_{k,m}W_{k,m}\log_2 h\left(g_{k,m}, \varepsilon_c\right), & \text{with } p = (1 - \varepsilon_c) \\ 0, & \text{with } p = \varepsilon_c \end{cases} \tag{24}$$

and

$$h\left(g_{k,m}, \varepsilon_c\right) = \begin{cases} u\left(g_{k,m}\right), & \text{with } p = (1 - \varepsilon_c) \\ 1, & \text{with } p = \varepsilon_c \end{cases} \tag{25}$$

where $u(g_{k,m}) = [(1 + Jg_{k,m})/(e^{\sqrt{V_{k,m}}N})]$, we explore the Mellin transform of $\psi_{k,m}$

$$\begin{aligned} \mathcal{M}_{\psi_{k,m}}(s) &= E\left[e^{r_{k,m}(s-1)}\right] \\ &= E\left[e^{\frac{B_{k,m}W_{k,m}\cdot(s-1)}{\ln 2}\ln h(g_{k,m},\varepsilon_c)}\right] \\ &= E\left[h\left(g_{k,m}, \varepsilon_c\right)^{\frac{B_{k,m}W_{k,m}\cdot(s-1)}{\ln 2}}\right] \\ &= (1 - \varepsilon_c)\cdot E\left[u\left(g_{k,m}\right)^{\frac{B_{k,m}W_{k,m}\cdot(s-1)}{\ln 2}}\right] + \varepsilon_c \\ &= (1 - \varepsilon_c)\cdot \mathcal{M}_{u(g_{k,m})}\left[1 + \frac{B_{k,m}W_{k,m}\cdot(s-1)}{\ln 2}\right] \\ &\quad + \varepsilon_c. \end{aligned} \tag{26}$$

The problem is converted to finding the Mellin transformation of $u(g_{k,m})$. In Rayleigh fading channel, the channel gain $g_{k,m}$ is a random variable which follows the exponential distribution of parameter $\overline{g}_{k,m}$ [34]:

$$\begin{aligned} \mathcal{M}_{u(g_{k,m})}(s) &= E\left[\left(\frac{1 + Jg_{k,m}}{e^{\sqrt{V_{k,m}}N}}\right)^{s-1}\right] \\ &= \int_0^{+\infty}\left(\frac{1 + Jg_{k,m}}{e^{\sqrt{V_{k,m}}N}}\right)^{s-1} e^{g_{k,m}}dg_{k,m}. \end{aligned} \tag{27}$$

Combining (19), (20), (26), and (27), we, finally, get the upper bound of the delay violation probability as (21), shown at the bottom of the page.

B. Problem Statement

We use the delay violation probability $\varepsilon^q_{k,m}$ and the decoding error probability ε_c to express the overall reliability ε as (28). When the reliability values of the $\varepsilon^q_{k,m}$ and ε_c are pretty small, the approximation of (28) can be proved to be very accurate [35]

$$\varepsilon_{k,m} = 1 - \left(1 - \varepsilon^q_{k,m}\right)(1 - \varepsilon_c) \approx \varepsilon^q_{k,m} + \varepsilon_c. \tag{28}$$

Refer to our previous work, the delay D for a IoTD scheduled in the downlink network is composed of five parts: queuing delay d_Q, frame alignment delay d_{FA}, transmission delay d_{Tx} and the processing delay of the BS d_{bsp} and mobile terminal d_{mtp}. The delay D can be expressed as (29). We still ignore the impact of d_{bsp} and d_{mtp} in this paper [29].

$$D = d_Q + d_{\text{FA}} + d_{\text{Tx}} + d_{\text{bsp}} + d_{\text{mtp}}. \tag{29}$$

Short TTI can reduce the delay of service, but at the same time it leads to a large control overhead, which reduces the SE of the network. PD transmission helps to improve the reliability, but redundant transmission obviously leads to a decrease of EE. How to improve EE and SE while reducing delay and improving reliability is an important issue that worth exploring.

The EE is defined as the ratio of bits successfully received by the IoTDs to the energy consumed by the BS, and the SE is defined as the ratio of the IoTD throughput to the system bandwidth. They can be expressed, respectively, as

$$\eta_{\text{EE}} = \frac{\sum_{k\in\mathcal{U}_\mathcal{K}}\mu_k}{\sum_{m\in\mathcal{B}_\mathcal{K}}\sum_{k\in\mathcal{U}_\mathcal{K}}P_{k,m}\cdot t_{k,m}} \tag{30}$$

$$\eta_{\text{SE}} = \frac{\sum_{m\in\mathcal{B}_\mathcal{K}}\sum_{k\in\mathcal{U}_\mathcal{K}}r_{k,m}}{N_{\text{BS}}\cdot W_{\text{BS}}} \tag{31}$$

where N_{BS} is the number of BSs in the system, and W_{BS} is the bandwidth of BS. $t_{k,m}$ represents the transmission time of IoTD k.

EE and SE in the system are often inconsistent or even contradictory [36]. Therefore, we apply the relevant knowledge of game theory to achieve the compromise between EE and SE. Refer to the Cobb–Douglas production function in economics [37], we regard utility function as production, EE and SE as labor and capital, respectively, and define a new measure O. In order to achieve the trade-off between EE and

$$\varepsilon^q_{k,m} = \inf_{s>0}\left\{\frac{\left[(1 - \varepsilon_c)\cdot\int_0^{+\infty}\left(\frac{1+Jg_{k,m}}{e^{\sqrt{V_{k,m}}N}}\right)^{\frac{B_{k,m}W_{k,m}\cdot(-s)}{\ln 2}}\cdot e^{-g_{k,m}}\cdot dg_{k,m} + \varepsilon_c\right]^{D^q_{\max}}}{1 - e^{\mu\lambda_k(e^s-1)}\cdot\left[(1 - \varepsilon_c)\int_0^{+\infty}\left(\frac{1+Jg_{k,m}}{e^{\sqrt{V_{k,m}}N}}\right)^{\frac{B_{k,m}W_{k,m}\cdot(-s)}{\ln 2}}\cdot e^{-g_{k,m}}\cdot dg_{k,m} + \varepsilon_c\right]}\right\} \tag{21}$$

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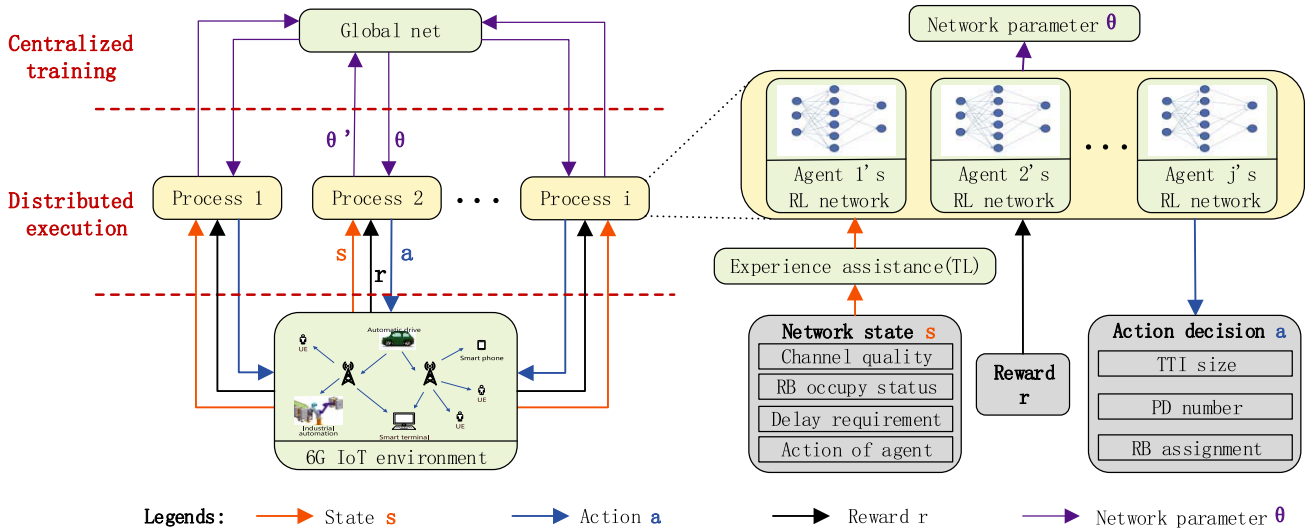

Fig. 3. Distributed TA3C scheme.

SE, we normalize EE and SE separately in (32) so that they are comparable to be integrated into one function

$$\eta_{\mathrm{EE}}^{\mathrm{norm}} = \frac{\eta_{\mathrm{EE}}}{\eta_{\mathrm{EE}}^{\max}}, \quad \eta_{\mathrm{SE}}^{\mathrm{norm}} = \frac{\eta_{\mathrm{SE}}}{\eta_{\mathrm{SE}}^{\max}}. \tag{32}$$

Assuming that the preference for EE $\vartheta \in [0, 1]$ is given by the system, then the optimization objective can be expressed as follows:

$$O\left(P_{k,m}, W_{k,m}, T_{k,m}^{f}, N_{k,m}^{\mathrm{PD}}\right) = \left(\eta_{\mathrm{EE}}^{\mathrm{norm}}\right)^{\vartheta}\left(\eta_{\mathrm{SE}}^{\mathrm{norm}}\right)^{1-\vartheta}. \tag{33}$$

Therefore, the optimization problem can be expressed as

$$\begin{aligned}
\max \quad & O\left(P_{k,m}, W_{k,m}, T_{k,m}^{f}, N_{k,m}^{\mathrm{PD}}\right) \\
\text{s.t.} \quad & C_{34.1} \quad (21), (29) \\
& C_{34.2} \quad D_k \le D_{\max}^{k} \\
& C_{34.3} \quad \left(\varepsilon_{k,m}^{q} + \varepsilon_c\right)^{N_{k,m}^{\mathrm{PD}}} < \varepsilon_{\max}^{k} \\
& C_{34.4} \quad \sum_{k \in \mathcal{U}_{\mathcal{K}}} P_{k,m} \le P_{\max} \\
& C_{34.5} \quad \sum_{k \in \mathcal{U}_{\mathcal{K}}} W_{k,m} \le W_{\max} \\
& C_{34.6} \quad N_{k,m}^{\mathrm{PD}} \in \{0, 1, 2, 3\} \\
& C_{34.7} \quad T_{k,m}^{f} \in \{0.0625 \text{ ms}, 0.125 \text{ ms}, 0.25 \text{ ms}\}.
\end{aligned} \tag{34}$$

Constraint $C_{34.1}$ are the calculations of delay violation probability and latency. Constraint $C_{34.2}$ indicates that the delay of IoTD is less than the maximum delay requirement, while $C_{34.3}$ represents the reliability requirements of IoTD, and PD significantly improves the reliability. $C_{34.4}$ and $C_{34.5}$ require that the power and bandwidth of the BS do not exceed the maximum value. $C_{34.6}$ is the range of the number of PD. $C_{34.7}$ represents the options of TTI.

For the convenience of simulation analyses and description, we introduce $\overline{\gamma}_{k,m}$ to represent the average value of SNR, and

$$\overline{\gamma}_{k,m} = \overline{g}_{k,m} \cdot \frac{\alpha P_{k,m}}{N_0 B_{k,m} W_{k,m}}. \tag{35}$$

Due to the complexity of SNC method and finite block-length coding, this optimization problem is difficult to solve with traditional optimization methods. Most of the existing work avoids this problem, or takes a lot of effort in the process of calculation and simplification [26], [38]. Therefore, we use ML method, which is good at solving complex decision problems, to achieve the cross-layer parameter optimization and resource allocation.

V. Solution With ML Methods

We aim at proposing an intelligent self-organizing solution to fulfill the delay and reliability requirements in the IoT scenarios and provide IoTDs with customized services. Specifically, we propose the TA3C algorithm to solve the joint TTI, PD and RB allocation problem in (34), which combines the ideas of TL and the A3C algorithm.

Combined with the Markov properties of wireless fading channels, we model the problem of IoTD scheduling and resource allocation as a discrete-time Markov decision process. Agents interact with the wireless network environment repeatedly to learn how to map the state of the environment to the best action. The goal of training is to maximize the sum of accumulated rewards. To make the proposed TA3C algorithm easier to be understood, we will introduce our algorithm in detail. The framework of TA3C algorithm is shown in Fig. 3.

Agent: Each IoTD in the IoT network.

Action Space: For the state at each slot, the actions that the agent can choose are introduced as $\mathcal{A}^L = \{a_{\mathrm{TTI}}, a_{\mathrm{PD}}, a_{\mathrm{RB}}\}$, where a_{TTI} is the TTI size of the agent, a_{PD} is the number of PD and a_{RB} is the number of RBs allocated to this agent from system resources.

State Space: The network state can be expressed as $\mathcal{S}^L = \{\varsigma_{ch}, \varsigma_{\mathrm{RB}}, \varsigma_{de}, \varsigma_{ac}\}$, where ς_{ch} indicates the channel quality, ς_{RB} is the situation of RB occupation, ς_{de} represents the delay requirement of each agent and ς_{ac} records the action selection of each agent at the last slot.

Reward Function: The reward function is a very important part of the entire ML network. Each agent achieves the goal of

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maximizing the reward value through continuous interaction with the IoT network environment. Due to the urgent requirements of green wireless networks, we pay more attentions to the trade-off of system EE and SE, and use the expression of (33) as the reward function.

We define state transition probability $P(s_{t+1}|s_t, a_t)$, which represents the probability of transferring from a state $s_t \in \mathcal{S}^{\mathcal{L}}$ to a state $s_{t+1} \in \mathcal{S}^{\mathcal{L}}$ after choosing an action $a_t \in \mathcal{A}^{\mathcal{L}}$. Moreover, we use r_t^L represents the immediate reward at the time slot t. $\beta \in [0, 1)$ is the discount factor. The state-action value function is given by $Q^{\pi}(s, a)$, it expresses the benefits of the policy π obtained when performing the action a at state s. The right side of (36) shows that it can also be calculated recursively using Bellman equation

$$\begin{aligned} Q^{\pi}(s, a) &= E\left[\sum_{t=1}^{\infty} \beta^t r_t^L(s_t, a_t)|s_0 = s, \pi\right] \\ &= E\left[r_t^L(s_t, a_t) + \beta Q^{\pi}(s_{t+1}, a_{t+1})\right]. \end{aligned} \tag{36}$$

$d(s)$ is used to represent the state distribution function under policy π [39]. The purpose of network training is to find a strategy to maximize the reward function, which is expressed as

$$\begin{aligned} J(\pi) &= E\left[Q^{\pi}(s, a)\right] \\ &= \int_{\mathcal{S}^{\mathcal{L}}} d(s) \int_{\mathcal{A}^{\mathcal{L}}} \pi(s, a) Q^{\pi}(b, a) dads. \end{aligned} \tag{37}$$

Common ML methods include value-based and policy-based methods. The A3C algorithm combines the value-based and policy-based methods, and it is composed of two parts: actor and critic. The actor selects an action of TTI, PD, and RB based on the observed IoT network state, while the critic is responsible for giving the evaluation of the strategy by the rewards.

A. Actor Process

The actor uses stochastic policy gradient method to update the parameterized policies. Then, the gradient of the policy is given by

$$\nabla_{\theta} J(\pi_{\theta}) = \int_{\mathcal{S}^{\mathcal{L}}} d(s) \int_{\mathcal{A}^{\mathcal{L}}} \nabla \pi_{\theta}(s, a) A(s, a) dads \tag{38}$$

where $d(s)$ denotes the state distribution function, $A(s, a) = Q_w(s, a) - Q_{\upsilon}(s)$ is the advantage function. We use the Gibbs distribution to express the parameterized stochastic policy $\pi_{\theta}(s, a)$

$$\pi_{\theta}(s, a) = \frac{e^{\theta^T \phi(s,a)}}{\sum_{a' \in \mathcal{A}^{\mathcal{L}}} e^{\theta^T \phi(s,a')}} \tag{39}$$

where $\phi(s, a)$ is the feature vector. We use σ_a represents the learning rate of actor. Then, the policy parameter vector is updated as

$$\theta_{t+1} = \theta_t + \sigma_a \nabla_{\theta} J(\pi_{\theta}). \tag{40}$$

B. Critic Process

Since Bellman equation cannot calculate Q for infinite state and action problems, we use linear combination function approximation to estimate the value function $Q_{\upsilon}(s)$ and state-action value function $Q_{\omega}(s, a)$

$$Q_{\upsilon}(s) = \upsilon^T \psi(s) \tag{41}$$

$$Q_w(s, a) = w^T \varphi(s, a) \tag{42}$$

where $\psi(s) = \{\varsigma_{ch}, \varsigma_{\mathrm{RB}}, \varsigma_{de}, \varsigma_{ac}\}$ is the features vector at state s, $\varphi(s, a) = (\varphi_1(s, a) \cdots \varphi_M(s, a))^T$ is the feature vector when the agent chooses the action a at the state s. w and υ are parameter vectors. Then, the temporal difference (TD) error between the estimated value and the real value can be estimated as

$$\delta_t^w = r_{t+1} + \beta Q_w(s_{t+1}, a_{t+1}) - Q_w(s_t, a_t). \tag{43}$$

An eligibility trace mechanism is adopted for more efficient and quality learning. Let z_t^w denote the eligibility trace vector, then, the update equations for the z_t^w and the critic parameter vector w_{t+1} can be expressed as

$$z_t^w = \kappa \beta z_{t-1}^w + \varphi(s, a) \tag{44}$$

$$w_{t+1} = w_t + \sigma_c \delta_t^w z_t^w \tag{45}$$

where $\kappa \in [0, 1)$ is the trace-decay parameter, σ_c is the learning rate of critic. Similarly, the parameters related to state features have the same forms as (43)–(45).

C. TA3C Algorithm

We summarize the principles of A3C in a concise way, and propose the TA3C algorithm based on the above theory. The specific algorithm is shown in Algorithm 1. In our intelligent framework, agents are encouraged to learn knowledge from other experienced agents by exchanging information. This is the inspiration given by TL.

At the same time, we use multiprocess parallel computing to improve the algorithm efficiency and performance. There is a main network, several worker processes, and each process has the same network as the public main network. Each process interacts with the environment independently to obtain experience data, and these processes learn independently and will not interfere with each other. After certain steps, each process synchronizes with the global net. The value function parameters and strategy parameters are updated in sequence. The values of the value function parameters w and υ are updated by the critics, and the strategy parameters are updated by the actors according to the evaluations of critics.

The TA3C algorithm can satisfy the constraints of (34) to realize the optimization problem, and the specific process is shown in Fig. 4. The constraints are realized through setting the value of reward. After the policy outputs the action, the agent executes the current action. First, whether the sum of bandwidth and power of IoTDs exceed the total network limitation is determined, corresponding to the restriction of $C_{34.4}$ and $C_{34.5}$. Then, calculate whether the delay is less than the maximum tolerance. Finally, the delay violation probability is calculated according to constraints $C_{34.1}$ and $C_{34.3}$. If the

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Algorithm 1 The Specific TA3C Algorithm

1: **Input:** Learning rate σ_c and σ_a, discount factor β, decay factor κ, IoT environment parameters.
2: **Initialize:** Initial state $s_0, Q_\omega(s,a), \pi_\theta(s,a), \theta_0, z_t^w, z_t^\upsilon, w_0, \upsilon_0$
3: **For** each episode $EP = 0, 1, 2, \ldots, EP_{\max}$ **do**
4: Initialize the network state s
5: **For** each step $t = 0, 1, 2, \ldots, T$ steps **do**
6: **if** there is records with similar experience
7: Get the transferred strategy from the past experience
8: Take action and update related parameter sets
9: Perform efficient AC learning from step 12 to step 21
10: Take action a_{t+1} according policy $\pi_\theta(s,a)$
11: observe the state s_{t+1} and immediate reward r_{t+1}
12: **Critic Process**
13: Complete the approximate of the function:
$Q_\upsilon(s) = \upsilon^T \psi(s)$
$Q_w(s,a) = w^T \varphi(s,a)$
14: Calculate TD error:
$\delta_t^w = r_{t+1} + \beta Q_w(s_{t+1}, a_{t+1}) - Q_w(s_t, a_t)$
$\delta_t^\upsilon = r_{t+1} + \beta Q_\upsilon(s_{t+1}) - Q_\upsilon(s_t)$
15: Update eligibility trace:
$z_t^w = \kappa\beta z_{t-1}^w + \varphi(s,a)$
$z_t^\upsilon = \kappa\beta z_{t-1}^\upsilon + \psi(s)$
16: Update critic parameters:
$w_{t+1} = w_t + \sigma_c \delta_t^w z_t^w$
$\upsilon_{t+1} = \upsilon_t + \sigma_c \delta_t^\upsilon z_t^\upsilon$
17: **Actor Process**
18: Calculate advantage function:
$A(s_{t+1}, a_{t+1}) = Q_w(s_{t+1}, a_{t+1}) - Q_\upsilon(s_{t+1})$
19: Update policy gradient:
$\nabla_\theta J(\pi_\theta) = \int_{\mathcal{S}^L} d(s) \int_{\mathcal{A}^L} \nabla\pi_\theta(s,a) A(s,a) dads$
20: Update policy parameters:
$\theta_{t+1} = \theta_t + \sigma_a \nabla_\theta J(\pi_\theta)$
21: Record the data of the episode
22: **End for**
23: **End for**
24: **End**

requirement of constraints are met, the reward is calculated by (33). Otherwise, the reward value is set to 0.

D. Advantages of TA3C Algorithm

The proposed TA3C algorithm has the following advantages.

1) Multiagent ML method is adopted in the TA3C. Agents share the same environment, perceive the environment and take actions. In our problem, different IoTDs share communication resources and influence each other. Multiagent system can increase the robustness and scalability of the system [40], but it will also make the dimensional disaster more serious. As the state and action space increase, the complexity of computation increases exponentially. To solve this problem, we use TL and distributed computation to reduce the time costs.
2) In order to solve the problem of complex calculation, distributed method is adopted in TA3C, which can operate in parallel to speed up the model convergence. Through the distributed execution and centralized training, each process interacts with the IoT network environment about the state space, action space and reward. Each process trains the ML network independently, and exchanges network parameters with the global net to maintain synchronization.
3) TL is to improve the learning performance of new tasks by transferring knowledge and experience from related tasks that have been learned. TL is a shortcut to save time and get better performance. Although ML and deep learning have powerful performance, they are often costly. They require massive amounts of data, expensive hardware facilities, and high engineer costs. When training a learning model requires huge computing resources and datasets, the advantages of TL are obvious. It does not need to develop a completely new solution, but to transfer knowledge from related problems more easily. In other words, TL does not need to start from "zero," but learn from the shoulders of giants. Specific to our work, after collecting the characteristic information of services, the network will first determine whether there is information about past experience to provide assistance. In the problem we analyzed, the TL brings more benefits in saving computation resources and reducing time cost.

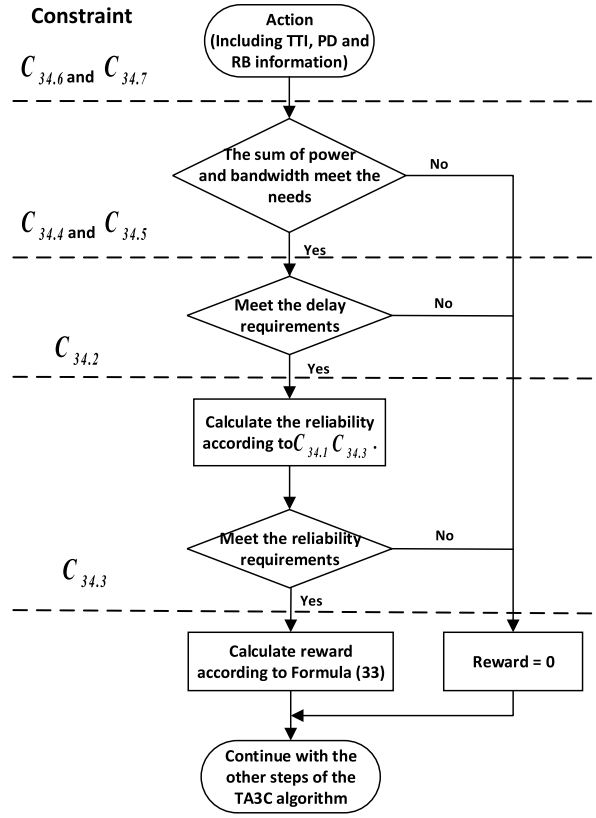

Fig. 4. Process of the TA3C algorithm realizing optimization problem.

VI. Simulation Results and Analyses

A. Simulation Parameter

Based on related 3GPP standards [41], [42], we perform the simulation on the Python platform. IoTDs are randomly distributed within the coverage area of the BSs. The service

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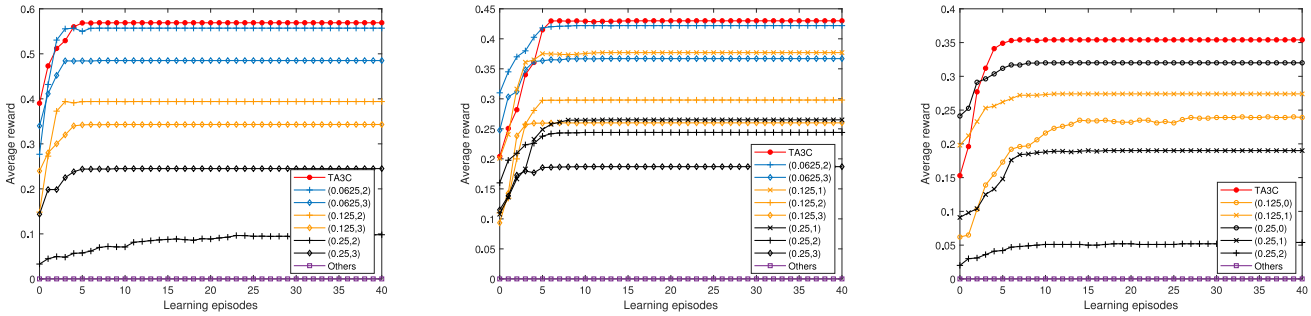

(a) (b) (c)

Fig. 5. Comparison of cross-layer optimization scheme with flexible parameter configuration and fixed parameter configuration. The parameters in the legend represent (TTI size, number of PD). (a) $D_{\max} = 1$ ms, $\varepsilon_{\max} = 10^{-5}$, $\overline{\gamma} = 25$ dB. (b) $D_{\max} = 2$ ms, $\varepsilon_{\max} = 10^{-4}$, $\overline{\gamma} = 15$ dB. (c) $D_{\max} = 3$ ms, $\varepsilon_{\max} = 10^{-3}$, $\overline{\gamma} = 10$ dB.

TABLE II
SIMULATION PARAMETERS

Parameter	Value
IoTD distribution	Evenly distributed
Bandwidth of BS	20 MHz
Power of BS	46 dBm
Power of noise	-95 dBm
Pathloss of BS	$128.1 + 37.6 \log d(km)$ dB
Traffic model	Poisson arrival process
Basic scheduling principle	FCFS
Packet size	256 bits [44]
Preference of EE	0.5
Decoding error probability	2×10^{-6} [43]
TTI size	Optional variable: 0.0625ms; 0.125ms; 0.25ms [28]
Number of PD	Optional variable: 1; 2; 3
Number of RB	Optional variable: Integer not exceeding BS resources

arrival processes follow the Poisson distribution, and we adopt the FCFS as the basic scheduling principle. The specific parameters are shown in Table II. In this article, we regard the decoding error probability as a fixed value [43].

When the service arrives at the BS, the BS refers to the service characteristics and network states, completes the flexible configuration and optimization of parameters, such as TTI, the number of PD, and RB resource occupation based on TA3C algorithm. The flexible scheduling, allocation of resources and customized services based on optimized parameters are realized. In this section, we simulate and analyze the performance of communication system, flexible configuration parameters, and the ML algorithms.

B. System Performance

First, we illustrate the improvement of system performance brought by flexible parameter configuration in Fig. 5. We compare our proposed flexible algorithm with fixed-number of PD and fixed-size TTI scheduling. The TTI parameters that can be selected are 0.0625 ms, 0.125 ms, 0.25 ms, and the parameters of PD number are 0, 1, 2, and 3. We arrange the two variables and simulate all the conditions. The situations that can meet the needs of IoTDs are drawn with different lines, and the remaining situations that cannot meet the needs of IoTDs are merged into "others." Different parameter configurations all use the TA3C algorithm for resource scheduling and allocation. Due to the normalization, the reward values are between 0 and 1, which will be more conducive to the convergence of our algorithm. In Fig. 5, we do not consider the difference between IoTDs, the delay and reliability requirements of IoTDs in the system are consistent.

The three graphs in Fig. 5 show that flexible parameter configuration in the proposed TA3C always has performance advantages under the different conditions of delay, reliability requirements and SNR. Comparing different pictures, it is obvious that different parameters should be selected in different situations. When the delay and reliability requirements are very strict as shown in Fig. 5(a), it is difficult to meet the reliability requirements of IoTDs without PD. At the same time, the short TTI has performance advantages to reduce the frame alignment delay and transmission delay, which has been proved in detail in our previous work. As the delay and reliability requirements become loose as shown in Fig. 5(b), more parameter configurations have the opportunity to meet IoTD needs, but there are differences in the performance that can be achieved. Still shorter TTI and duplication transmission have performance advantages. But when the delay requirement becomes 3 ms and the reliability requirement is 10^{-3}, as shown in Fig. 5(c), we observe that the parameter configuration with performance advantages has changed. This is because although the PD and shorter TTI help to meet the strict requirements, it will lead to a decrease in the SE and EE of the network. On the other hand, due to the poor channel quality, and short TTI will inevitably lead to a short data transmission time T_t, the delay violation probability becomes high as shown in (21). Therefore, 0.0625-ms TTI cannot meet the needs of IoTDs even if PD is used. In addition, when the parameters are flexibly configured, the action space of the learning algorithm is larger, that is, more learning dimensions are required, so it may take longer steps to converge.

Different from Fig. 5, Fig. 6 fully considers the differences between IoTDs, different IoTDs in the system have different

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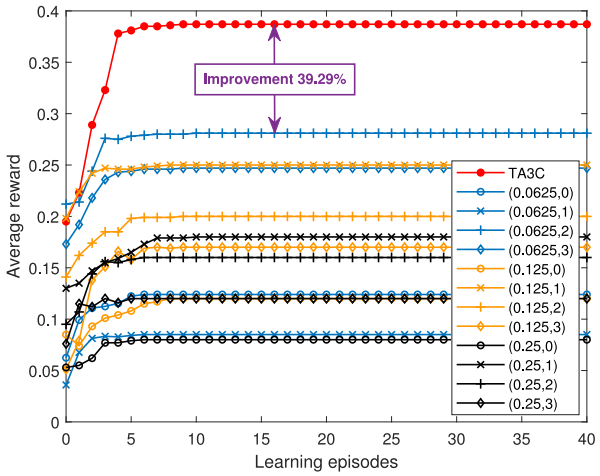

Fig. 6. Performance comparison between TA3C algorithm and fixed parameter configuration considering IoTD differences. The parameters in the legend represent (TTI size, number of PD). $\overline{\gamma} = 15$ dB.

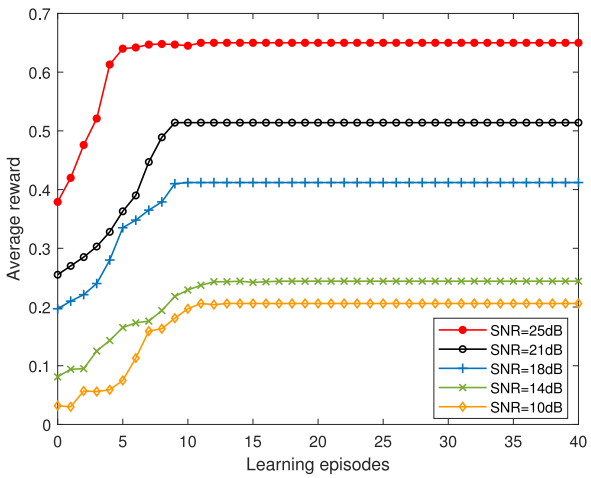

Fig. 7. Impact of SNR on system performance.

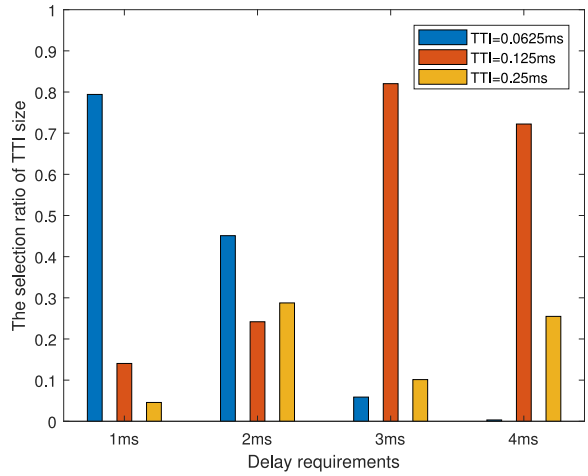

Fig. 8. Impact of IoTD delay requirements on TTI selection ratio. $\overline{\gamma} = 15$ dB.

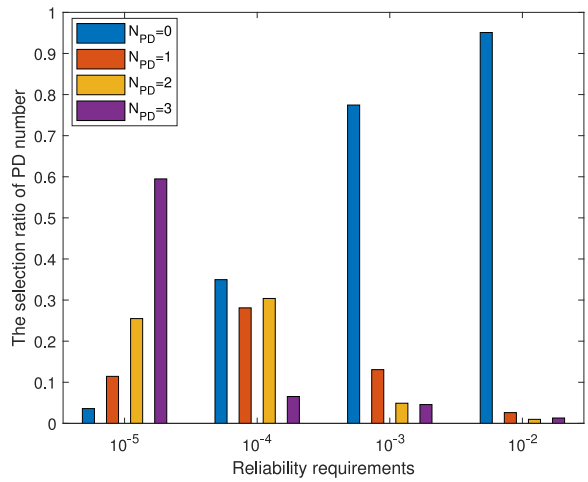

Fig. 9. Impact of IoTD reliability requirements on PD number selection. $\overline{\gamma} = 15$ dB.

delay and reliability requirements. It is more in line with the actual network requirements, and can reflects the network flexibility and customized service ability. In our simulation, the IoTD delay requirements are distributed in the range of 1–4 ms, and the reliability requirements are distributed in the range of 10^{-5} to 10^{-2}. Fig. 6 shows the comparison between our proposed TA3C algorithm and the fixed parameter configuration. Compared with Fig. 5, we can notice that the advantage of flexible parameter configuration is more obvious when there are differences in IoTD requirements, and the performance is improved by 39.29% compared with the best performance of fixed configuration.

Fig. 7 shows the impact of SNR on system performance. In the simulation, the IoTD delay and reliability requirements are randomly distributed. With the improvement of channel quality, the trade-off values of SE and EE increase, which is consistent with our theoretical analyses. In addition, we observe that when the channel quality is good, the algorithm converges faster. This is because when the SNR is large, the reliability that can be achieved become larger, and the algorithm is easy to learn good performance.

C. Flexible Configuration

The effect of delay requirements on TTI selection is shown in Fig. 8. Our simulation is completed in the environment of SNR = 15 dB, and the reliability requirements are randomly distributed in the range of 10^{-5} to 10^{-2}. We can notice that the delay requirement has a great influence on the choice of TTI. With the decrease of delay requirement, the proportion of 0.0625 ms decreases obviously, and larger TTI has advantages. However, the TTI of 0.25 ms will lead to a large frame alignment delay. Thus, TTI of 0.25 ms is not very popular in most cases. The flexible parameter configuration scheme balances these relationships and keeps the system performance optimal under various conditions.

The change of PD number with reliability requirement is analyzed in Fig. 9. When the reliability requirement is 10^{-5}, most IoTDs choose to duplicate the packet twice or even three times. As the demand of reliability becomes looser, the proportion and frequency of duplication become less. When the reliability is 10^{-4}, several cases have similar performance. When the reliability requirement decreases to 10^{-2}, most IoTDs no longer need to duplication transmission. From the

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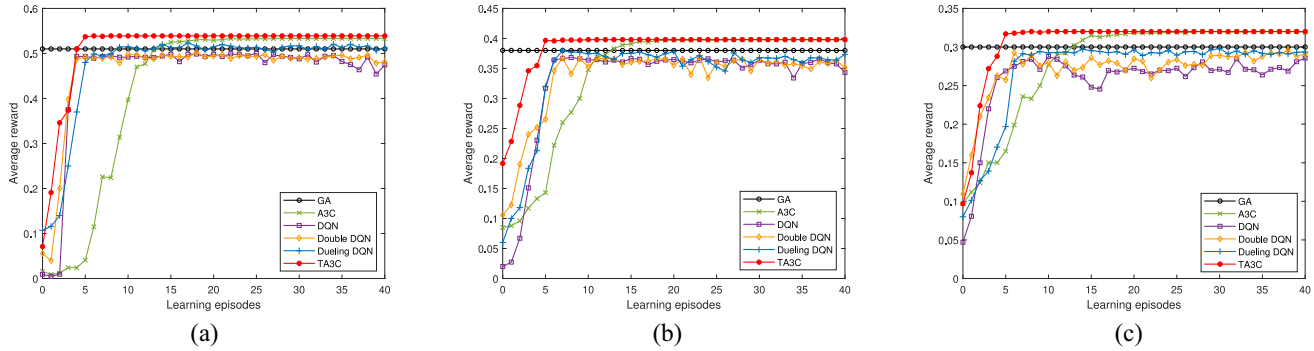

Fig. 10. Performance comparison between TA3C algorithm and other algorithms. (a) $D_{\max} = 1$ ms, $\varepsilon_{\max} = 10^{-5}$, $\overline{\gamma} = 23$ dB. (b) $D_{\max} = 2$ ms, $\varepsilon_{\max} = 10^{-4}$, $\overline{\gamma} = 13$ dB. (c) $D_{\max} = 3$ ms, $\varepsilon_{\max} = 10^{-3}$, $\overline{\gamma} = 8$ dB.

analyses of Fig. 9, it is not difficult to draw a conclusion that PD transmission is very helpful to improve the reliability performance.

D. Algorithm Performance

We compare the proposed TA3C algorithm with other intelligent and traditional methods. The TA3C algorithm shows performance advantages not only in system performance but also in operating efficiency and stability. The specific comparison algorithms are as follows.

1) DQN: The value-based algorithm represented by deep Q-learning (DQN) is the most common ML algorithm [40]. It combines the neural network and Q-learning. Due to the limited ability of function approximation, the DQN algorithm will overestimate the value of actions under certain conditions.

2) Double DQN: Double DQN solves the problem of overestimation by decoupling action selection and action value evaluation [45]. Double DQN has two Q-network structures. Two value functions are trained equally by randomly sampling the experience of the agents.

3) Dueling DQN: Dueling DQN algorithm improves the model structure based on DQN. The whole model structure is divided into two parts, that is, state value function and the advantage function. Dueling DQN can estimate the value function more accurately and has better algorithm performance.

4) A3C: A3C algorithm includes action module actor and evaluation module critic [39]. The actor outputs action according to the input state. Critic adjusts itself according to the historical information and environment feedback, and then commands the actor. A3C algorithm can solve both the discrete and the continuous space problem. It is a widely used ML algorithm.

5) GA: Genetic algorithm (GA) is a method of searching for the optimal solution by simulating the natural evolution process. Compared with other conventional optimization algorithms, GA usually has better optimization performance when solving more complex and combinatorial optimization problems.

Then, we analyze the performance of our proposed algorithm compared with DQN, double DQN, dueling DQN, A3C, and GA in Fig. 10. Our comparison methods include the common methods suitable for discrete action spaces. Because GA and ML algorithms are expressed in different ways and cannot be displayed in the same coordinate axis, we only show the final result of GA after convergence with a horizontal line. The three pictures are completed with different simulation parameter settings. It is obvious that the TA3C algorithm always has the best performance. It not only converges to the point where the reward is larger, but also has the most stable performance. After converging, comparison algorithms still have large oscillations. Without considering the difference of IoTDs, A3C algorithm can converge to the same performance as TA3C, but has a slower convergence speed.

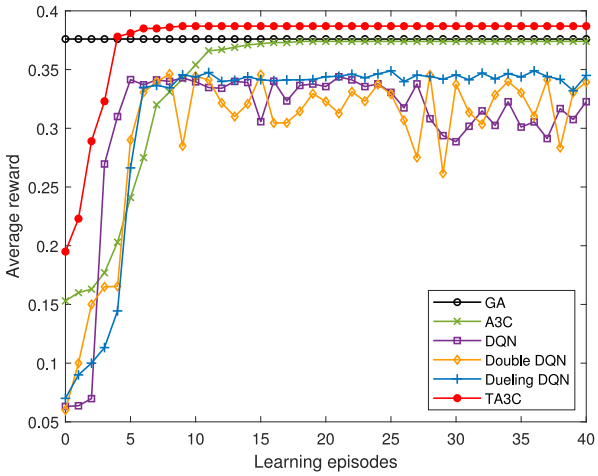

Fig. 11. Performance comparison of different algorithms considering IoTD differences. $\overline{\gamma} = 15$ dB.

Fig. 11 compares the performance differences of different algorithms when considering IoTD differences. We notice that TA3C algorithm still has good convergence performance. The convergence performance of A3C algorithm becomes worse, and it no longer has the same performance as TA3C algorithm. At this case, the performance of DQN and double DQN algorithm are very unstable after convergence, and the oscillation is more obvious. This is because comparison algorithms do not consider the difference between agents. The performance of dueling DQN is more stable compared with DQN and double DQN, but it does not converge to the optimal value. When the requirements of IoTDs are different, the convergence performance of the algorithm decreases.

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TABLE III
COMPLEXITY AND RELIABILITY OF DIFFERENT ALGORITHMS

Different algorithms	Learning time(s)	Algorithm reliability
DQN	7989.27	84%
Double DQN	8573.57	86%
DuelingDQN	7266.36	94%
A3C	3001.14	93%
TA3C	263.41	98%

Table III shows the comparisons of algorithm complexity and reliability. We use the algorithm running time to characterize the algorithm complexity. Our simulation is accomplished on the same computer with same parameters. We do 50 episodes at one time, 300 steps per episode. Each algorithm executed ten times to get the average value. The complexities of DQN, double DQN, and dueling DQN algorithms are similar, while the A3C algorithm takes less time because of its distributed learning method. Due to the introduction of TL, the advantages of TA3C algorithm are very obvious. Compared with A3C algorithm, the complexity of the TA3C algorithm is reduced about 91.23%.

Then, we analyze the reliability of the algorithms. A cross-referencing method is proposed in [46] to test the reliability of the ML algorithms. The same input is tested repeatedly and the inconsistent convergence results are defined as failure. Algorithm reliability R_a is defined as [47]

$$R_a = 1 - \frac{F}{M} \tag{46}$$

where F represents the number of failed tests, and M is the number of total tests. Therefore, we test several ML algorithms with the same parameter environment repeatedly. Due to the limitation of complexity, DQN, double DQN, and dueling DQN algorithms are tested 50 times, A3C and TA3C algorithms are tested 100 times, and the results are shown in Table III. Simulation results show that the proposed TA3C has better reliability performance than other compared algorithms.

Futhermore, the influence of learning rate on the performance of TA3C algorithm is given in Fig. 12. We notice that when the learning rate is as large as 0.01, the system will converge quickly. However, it often leads to premature convergence, and the optimal value cannot be achieved. On the other hand, when the learning rate is too small, the convergence speed becomes too slow, the convergence process oscillates obviously and the performance is unstable. Therefore, choosing an appropriate learning rate is also important to apply ML methods.

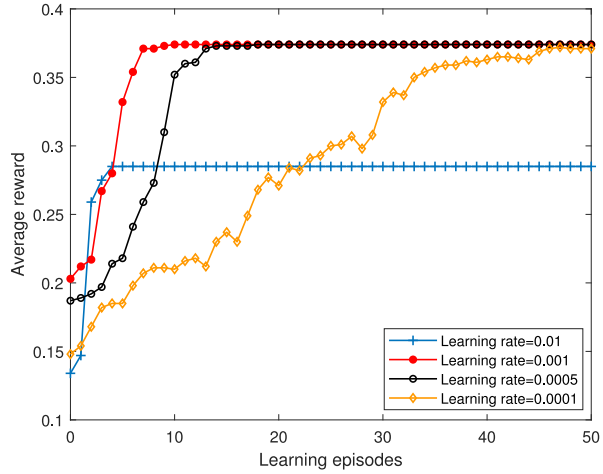

Fig. 12. Impact of different learning rates on the performance of TA3C algorithm.

VII. CONCLUSION

Combined with the concepts of green communication, flexible communication and intelligent communication, a cross-layer optimization framework is proposed in this article. The delay and reliability sensitive services in IoT scenarios are concerned, and a flexible parameter configuration and resource allocation scheme is proposed, which realizes the compromise and improvement of EE and SE. Flexible TTI scheduling and PD transmission are adopted to reduce delay and improve reliability. The TA3C algorithm is given to realize the flexible configuration of TTI, PD number, and RB allocation. The simulation results show that the performance of the proposed cross-layer optimization scheme is improved by at least 39.29%, and the TA3C algorithm has better convergence performance and reduces the complexity of algorithm by 91.23%.

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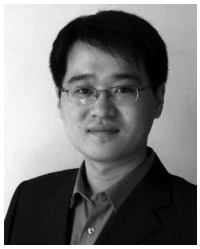

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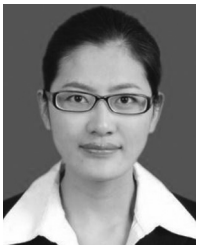

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